Unit: Unit 6: Systems of Linear Equations and Inequalities
Topic: Solving Systems by Graphing

Name: ________________
Date: ____________

Homework (Carolina Reaper)

Q1. A music festival has two stages with sound systems. The equations $2x+3y=12$ and $x-2y=-3$ represent the sound frequencies. What is the point of intersection?
(A) (3, 2)
(B) (2, 3)
(C) (1, 4)
(D) (4, 1)
(E) (0, 0)

Q2. An artist is creating a painting with a ratio of red to blue paint. The system $x+2y=8$ and $3x-2y=4$ represents the paint mixture. What is the ratio of red to blue paint?
(A) 2:1
(B) 1:2
(C) 3:2
(D) 2:3
(E) 1:1

Q3. A rhythm pattern has a system of equations $y=2x-1$ and $y=-x+3$. What is the intersection point of the two lines?
(A) (2, 3)
(B) (1, 2)
(C) (1, 1)
(D) (0, 1)
(E) (2, 1)

Q4. A music producer is mixing two audio tracks. The system $3x+2y=12$ and $x-3y=-4$ represents the audio frequencies. What is the solution to the system?
(A) (2, 1)
(B) (1, 2)
(C) (4, 0)
(D) (0, 4)
(E) (1, 1)

Q5. An art gallery has a system of equations $2x+y=7$ and $x-2y=-3$. What is the point of intersection?
(A) (1, 3)
(B) (2, 2)
(C) (3, 1)
(D) (4, 0)
(E) (0, 4)

Q6. A sound engineer is setting up a sound system. The equations $y=x+2$ and $y=-2x-1$ represent the sound frequencies. What is the point of intersection?
(A) (-1, 1)
(B) (1, 1)
(C) (0, 0)
(D) (1, 3)
(E) (-1, -3)

Q7. A painter is creating a mural with a system of equations $x+y=5$ and $2x-2y=2$. What is the point of intersection?
(A) (2, 1)
(B) (1, 2)
(C) (3, 1)
(D) (1, 4)
(E) (2, 3)

Q8. A music teacher is creating a rhythm pattern. The system $y=3x-2$ and $y=-x+1$ represents the rhythm. What is the intersection point of the two lines?
(A) (1, 1)
(B) (0, 1)
(C) (1, 0)
(D) (2, 1)
(E) (0, 0)

Open Ended Questions (FRQ)

Q9. Solve the system of equations $2x+3y=7$ and $x-2y=-1$ graphically and find the point of intersection. Provide your solution in the space below.

Q10. A sound system has a system of equations $y=2x-3$ and $y=-x+2$. Solve the system graphically and interpret the solution in the context of sound frequencies. Provide your solution in the space below.

Chef's Tips

- Tip 1: Emphasize the importance of accurate graphing
- Tip 2: Encourage students to check their solutions
- Tip 3: Provide feedback on interpretation of graphical solutions